

## 9 MICROBIOME SEQUENCING SERVICES MARKET, BY END USER

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### KEY FINDINGS

- The academic and research institutes segment accounted for the largest share of 45.0% of the global microbiome sequencing services market in 2022. This segment is projected to reach USD 247.53 million by 2028 from USD 127.82 million in 2023, at a CAGR of 14.1%.
- The pharmaceutical and biotechnology companies segment is expected to grow at the highest CAGR of 14.9% from 2023 to 2028.
- Other end users, including small-scale CROs, CDMOs, hospitals, and clinics, collectively accounted for a share of 23.0% of the global microbiome sequencing services market in 2022.

## 9.1 INTRODUCTION

Based on end user, the global microbiome sequencing services market is segmented into pharmaceutical and biotechnology companies, academic and research institutes, and other end users (including small-scale CROs, CDMOs, hospitals, and clinics). The academic and research institutes segment accounted for the largest market share of 45.0% in 2022. This can be attributed to the high adoption of microbiome sequencing services and clinical research in these facilities.

**TABLE 71** MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

| End User                                   | 2021          | 2022          | 2023          | 2024          | 2025          | 2026          | 2027          | 2028          | CAGR<br>(2023–2028) |
|--------------------------------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------------|
| Pharmaceutical and Biotechnology Companies | 70.17         | 80.20         | 91.62         | 104.76        | 119.89        | 137.69        | 158.69        | 183.68        | 14.9%               |
| Academic and Research Institutes           | 99.25         | 112.66        | 127.82        | 145.15        | 164.97        | 188.15        | 215.34        | 247.53        | 14.1%               |
| Other End Users                            | 50.62         | 57.31         | 64.87         | 73.48         | 83.30         | 94.76         | 108.18        | 124.03        | 13.8%               |
| <b>Total</b>                               | <b>220.03</b> | <b>250.17</b> | <b>284.31</b> | <b>323.39</b> | <b>368.16</b> | <b>420.61</b> | <b>482.21</b> | <b>555.24</b> | <b>14.3%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

## 9.2 PHARMACEUTICAL AND BIOTECHNOLOGY COMPANIES

### 9.2.1 ROBUST FUNDING FOR MICROBIOME-BASED THERAPEUTICS TO DRIVE MARKET

This segment is inclusive of the application of microbiome sequencing services in drug discovery, development, and clinical diagnostics. The microbiota plays an important role in the efficacy of therapeutic compounds in drug discovery. Many therapeutic compounds are processed by the catalytic functions of enzymes encoded by mammalian symbiotic bacteria. The unique nature of the human gut microbiome offers a new source of biomarkers to distinguish between patients' physiology and diseases for enhancement in personalized medicine.

Several diseases and conditions, such as diabetes, obesity, cancer, and autoimmune disorders, have been proposed to be treatable with the help of microbiome-based products. Several animal studies, preliminary human studies, uncontrolled studies, and anecdotal observations have shown positive results for these products. For instance, human trials indicating *Lactobacillus* GG may be useful in alleviating joint pains among rheumatoid arthritis (an autoimmune disease) patients. In addition, there are several animal studies that show that probiotics inhibit the initiation or progression of colon and bladder cancers. Furthermore, in vitro, cell culture, and animal studies have indicated that probiotics bind and prevent the absorption of aflatoxins, which have been associated with the occurrence of liver cancer in humans. This has created a significant demand for microbiome sequencing services among pharmaceutical and biotechnology companies.

The availability of funding for research is a key factor driving market growth, considering the rising pace of activity in this field. In the last five years, the NIH has extended funding of around USD 58.3 million to boost research in the human microbiome area, including funding for developing vaccines and devices. Of this, USD 49.0 million was marked for developing therapeutic products like prebiotics, probiotics, and dietary supplements and microbiome-based novel antimicrobials (Source: NIH).

## Recent developments in this market include:

- The Frontiers in Microbiology Journal article titled 'Key Technologies for Progressing Discovery of Microbiome-based Medicines,' published in June 2021, reported that human gastrointestinal microbiome research can deliver critical clinical and therapeutic development. It also suggested future progress dependent on effective links between sequencing approaches, computational analyses, and experimental assays.
- In July 2021, Seres Therapeutics entered into an agreement with Nestlé Health Science to jointly commercialize SER-109, Seres' investigational oral microbiome therapeutic for recurrent *Clostridioides difficile* infection (CDI) in the United States and Canada.

**TABLE 72** MICROBIOME SEQUENCING SERVICES MARKET FOR PHARMACEUTICAL AND BIOTECHNOLOGY COMPANIES, BY REGION, 2021–2028 (USD MILLION)

| Region                 | 2021         | 2022         | 2023         | 2024          | 2025          | 2026          | 2027          | 2028          | CAGR<br>(2023–2028) |
|------------------------|--------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------|---------------------|
| North America          | 40.30        | 45.71        | 51.82        | 58.80         | 66.77         | 76.09         | 87.02         | 99.94         | 14.0%               |
| Europe                 | 21.68        | 24.91        | 28.62        | 32.90         | 37.85         | 43.70         | 50.62         | 58.90         | 15.5%               |
| Asia Pacific           | 5.92         | 7.03         | 8.33         | 9.86          | 11.68         | 13.87         | 16.52         | 19.74         | 18.8%               |
| Latin America          | 1.34         | 1.51         | 1.71         | 1.92          | 2.16          | 2.44          | 2.76          | 3.13          | 12.9%               |
| Middle East and Africa | 0.93         | 1.04         | 1.15         | 1.28          | 1.43          | 1.59          | 1.77          | 1.97          | 11.3%               |
| <b>Total</b>           | <b>70.17</b> | <b>80.20</b> | <b>91.62</b> | <b>104.76</b> | <b>119.89</b> | <b>137.69</b> | <b>158.69</b> | <b>183.68</b> | <b>14.9%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

**TABLE 73** NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR PHARMACEUTICAL AND BIOTECHNOLOGY COMPANIES, BY COUNTRY, 2021–2028 (USD MILLION)

| Country      | 2021         | 2022         | 2023         | 2024         | 2025         | 2026         | 2027         | 2028         | CAGR<br>(2023–2028) |
|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------------|
| US           | 36.53        | 41.46        | 47.03        | 53.40        | 60.69        | 69.20        | 79.19        | 91.01        | 14.1%               |
| Canada       | 3.77         | 4.25         | 4.79         | 5.39         | 6.09         | 6.89         | 7.83         | 8.93         | 13.3%               |
| <b>Total</b> | <b>40.30</b> | <b>45.71</b> | <b>51.82</b> | <b>58.80</b> | <b>66.77</b> | <b>76.09</b> | <b>87.02</b> | <b>99.94</b> | <b>14.0%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

**TABLE 74** EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR PHARMACEUTICAL AND BIOTECHNOLOGY COMPANIES, BY COUNTRY, 2021–2028 (USD MILLION)

| Country/Region | 2021         | 2022         | 2023         | 2024         | 2025         | 2026         | 2027         | 2028         | CAGR<br>(2023–2028) |
|----------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------------|
| Germany        | 4.71         | 5.41         | 6.22         | 7.15         | 8.22         | 9.49         | 11.00        | 12.80        | 15.5%               |
| UK             | 4.21         | 4.86         | 5.62         | 6.50         | 7.52         | 8.73         | 10.17        | 11.90        | 16.2%               |
| France         | 3.32         | 3.81         | 4.38         | 5.04         | 5.79         | 6.69         | 7.75         | 9.02         | 15.5%               |
| Italy          | 0.93         | 1.07         | 1.23         | 1.41         | 1.62         | 1.87         | 2.16         | 2.51         | 15.3%               |
| Spain          | 0.72         | 0.82         | 0.94         | 1.08         | 1.24         | 1.42         | 1.65         | 1.91         | 15.2%               |
| Rest of Europe | 7.79         | 8.93         | 10.23        | 11.73        | 13.46        | 15.50        | 17.91        | 20.77        | 15.2%               |
| <b>Total</b>   | <b>21.68</b> | <b>24.91</b> | <b>28.62</b> | <b>32.90</b> | <b>37.85</b> | <b>43.70</b> | <b>50.62</b> | <b>58.90</b> | <b>15.5%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

**TABLE 75** ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR PHARMACEUTICAL AND BIOTECHNOLOGY COMPANIES, BY COUNTRY, 2021–2028 (USD MILLION)

| Country/Region       | 2021        | 2022        | 2023        | 2024        | 2025         | 2026         | 2027         | 2028         | CAGR<br>(2023–2028) |
|----------------------|-------------|-------------|-------------|-------------|--------------|--------------|--------------|--------------|---------------------|
| China                | 2.51        | 2.98        | 3.52        | 4.16        | 4.93         | 5.84         | 6.95         | 8.30         | 18.7%               |
| Japan                | 1.17        | 1.39        | 1.66        | 1.97        | 2.35         | 2.80         | 3.36         | 4.03         | 19.5%               |
| India                | 0.15        | 0.18        | 0.21        | 0.25        | 0.29         | 0.35         | 0.41         | 0.49         | 18.7%               |
| Rest of Asia Pacific | 2.10        | 2.49        | 2.94        | 3.48        | 4.11         | 4.88         | 5.80         | 6.92         | 18.7%               |
| <b>Total</b>         | <b>5.92</b> | <b>7.03</b> | <b>8.33</b> | <b>9.86</b> | <b>11.68</b> | <b>13.87</b> | <b>16.52</b> | <b>19.74</b> | <b>18.8%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

**TABLE 76** LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR PHARMACEUTICAL AND BIOTECHNOLOGY COMPANIES, BY COUNTRY, 2021–2028 (USD MILLION)

| Country/Region        | 2021        | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | 2028        | CAGR<br>(2023–2028) |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| Brazil                | 0.63        | 0.71        | 0.80        | 0.90        | 1.02        | 1.15        | 1.30        | 1.47        | 12.9%               |
| Rest of Latin America | 0.71        | 0.80        | 0.90        | 1.02        | 1.14        | 1.29        | 1.46        | 1.66        | 12.9%               |
| <b>Total</b>          | <b>1.34</b> | <b>1.51</b> | <b>1.71</b> | <b>1.92</b> | <b>2.16</b> | <b>2.44</b> | <b>2.76</b> | <b>3.13</b> | <b>12.9%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

## 9.3 ACADEMIC AND RESEARCH INSTITUTES

### 9.3.1 EXPANDING APPLICATIONS OF MICROBIOME RESEARCH TO DRIVE GROWTH

This segment is inclusive of the application of microbiome sequencing services in environmental and ecological research, veterinary research, and other research activities conducted by academic and research institutes.

Metagenomics is used in microbial ecology and environmental applications to study a wide range of microbial communities present in the physical surroundings (which cannot be cultivated in laboratories). Studying microbes in the environment helps to understand their potential use in various fields, such as agriculture and biotechnology. It also provides valuable insights into other areas, such as the ecology of water bodies, debris filtered from the air, or a sample of soil, to identify the species present in the environment. It can also play a crucial role in identifying a broad range of invasive and endangered species and tracking seasonal populations and variations.

The development of NGS techniques for the large-scale analysis of microbial communities and the rising number of ecological and environmental studies conducted across the globe will support the continued demand for microbiome sequencing in this end-user segment.

Advancements in DNA sequencing methodologies and data analysis have revolutionized research in many areas of veterinary sciences, including medicine, livestock management, and disease management. The veterinary applications of metagenomics include the identification of the complex consortium of bacteria, protozoa, and fungi, among others, and their interaction. The need for rapid and accurate diagnosis of pathogens in livestock species and changes in animal husbandry—including growing herd sizes—are likely to propel the market growth in this segment.

**TABLE 77** MICROBIOME SEQUENCING SERVICES MARKET FOR ACADEMIC AND RESEARCH INSTITUTES, BY REGION, 2021–2028 (USD MILLION)

| Region                 | 2021         | 2022          | 2023          | 2024          | 2025          | 2026          | 2027          | 2028          | CAGR<br>(2023–2028) |
|------------------------|--------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------------|
| North America          | 56.62        | 64.07         | 72.45         | 82.01         | 92.90         | 105.62        | 120.48        | 138.04        | 13.8%               |
| Europe                 | 30.81        | 35.05         | 39.86         | 45.38         | 51.70         | 59.10         | 67.81         | 78.13         | 14.4%               |
| Asia Pacific           | 8.52         | 9.93          | 11.55         | 13.44         | 15.64         | 18.25         | 21.36         | 25.09         | 16.8%               |
| Latin America          | 1.95         | 2.15          | 2.36          | 2.60          | 2.86          | 3.15          | 3.48          | 3.87          | 10.4%               |
| Middle East and Africa | 1.35         | 1.47          | 1.59          | 1.73          | 1.87          | 2.03          | 2.21          | 2.41          | 8.6%                |
| <b>Total</b>           | <b>99.25</b> | <b>112.66</b> | <b>127.82</b> | <b>145.15</b> | <b>164.97</b> | <b>188.15</b> | <b>215.34</b> | <b>247.53</b> | <b>14.1%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

**TABLE 78** NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR ACADEMIC AND RESEARCH INSTITUTES, BY COUNTRY, 2021–2028 (USD MILLION)

| Country      | 2021         | 2022         | 2023         | 2024         | 2025         | 2026          | 2027          | 2028          | CAGR<br>(2023–2028) |
|--------------|--------------|--------------|--------------|--------------|--------------|---------------|---------------|---------------|---------------------|
| US           | 51.31        | 58.11        | 65.77        | 74.50        | 84.47        | 96.10         | 109.72        | 125.81        | 13.9%               |
| Canada       | 5.31         | 5.96         | 6.69         | 7.51         | 8.44         | 9.51          | 10.76         | 12.23         | 12.8%               |
| <b>Total</b> | <b>56.62</b> | <b>64.07</b> | <b>72.45</b> | <b>82.01</b> | <b>92.90</b> | <b>105.62</b> | <b>120.48</b> | <b>138.04</b> | <b>13.8%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

**TABLE 79** EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR ACADEMIC AND RESEARCH INSTITUTES, BY COUNTRY, 2021–2028 (USD MILLION)

| Country/Region | 2021         | 2022         | 2023         | 2024         | 2025         | 2026         | 2027         | 2028         | CAGR<br>(2023–2028) |
|----------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------------|
| Germany        | 6.65         | 7.60         | 8.68         | 9.92         | 11.35        | 13.03        | 15.01        | 17.37        | 14.9%               |
| UK             | 5.95         | 6.83         | 7.84         | 8.99         | 10.33        | 11.91        | 13.77        | 16.00        | 15.3%               |
| France         | 4.71         | 5.36         | 6.10         | 6.95         | 7.93         | 9.07         | 10.42        | 12.02        | 14.5%               |
| Italy          | 1.33         | 1.51         | 1.71         | 1.94         | 2.21         | 2.52         | 2.88         | 3.31         | 14.1%               |
| Spain          | 1.02         | 1.16         | 1.31         | 1.48         | 1.68         | 1.91         | 2.18         | 2.50         | 13.8%               |
| Rest of Europe | 11.14        | 12.59        | 14.22        | 16.08        | 18.20        | 20.66        | 23.55        | 26.94        | 13.6%               |
| <b>Total</b>   | <b>30.81</b> | <b>35.05</b> | <b>39.86</b> | <b>45.38</b> | <b>51.70</b> | <b>59.10</b> | <b>67.81</b> | <b>78.13</b> | <b>14.4%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

**TABLE 80** ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR ACADEMIC AND RESEARCH INSTITUTES, BY COUNTRY, 2021–2028 (USD MILLION)

| Country/Region       | 2021        | 2022        | 2023         | 2024         | 2025         | 2026         | 2027         | 2028         | CAGR<br>(2023–2028) |
|----------------------|-------------|-------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------------|
| China                | 3.60        | 4.20        | 4.89         | 5.69         | 6.63         | 7.75         | 9.07         | 10.67        | 16.9%               |
| Japan                | 1.68        | 1.97        | 2.30         | 2.69         | 3.15         | 3.70         | 4.35         | 5.14         | 17.4%               |
| India                | 0.21        | 0.25        | 0.29         | 0.33         | 0.39         | 0.45         | 0.53         | 0.62         | 16.5%               |
| Rest of Asia Pacific | 3.03        | 3.52        | 4.07         | 4.72         | 5.47         | 6.36         | 7.41         | 8.66         | 16.3%               |
| <b>Total</b>         | <b>8.52</b> | <b>9.93</b> | <b>11.55</b> | <b>13.44</b> | <b>15.64</b> | <b>18.25</b> | <b>21.36</b> | <b>25.09</b> | <b>16.8%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

**TABLE 81**     **LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR ACADEMIC AND RESEARCH INSTITUTES, BY COUNTRY, 2021–2028 (USD MILLION)**

| Country/Region               | 2021        | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | 2028        | CAGR<br>(2023–2028) |
|------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| <b>Brazil</b>                | 0.91        | 1.01        | 1.11        | 1.22        | 1.35        | 1.49        | 1.65        | 1.83        | 10.5%               |
| <b>Rest of Latin America</b> | 1.03        | 1.14        | 1.25        | 1.37        | 1.51        | 1.66        | 1.84        | 2.04        | 10.3%               |
| <b>Total</b>                 | <b>1.95</b> | <b>2.15</b> | <b>2.36</b> | <b>2.60</b> | <b>2.86</b> | <b>3.15</b> | <b>3.48</b> | <b>3.87</b> | <b>10.4%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

## 9.4 OTHER END USERS

This segment is inclusive of the application of microbiome sequencing services in clinical research, diagnostics, and collaborative R&D activities conducted by small-scale CROs, CDMOs, hospitals, and clinics. Biomarkers provided by the human gut microbiome may allow a new means to distinguish between a patient's physiology and disease conditions. It can also enable mapping microbial fingerprints and defining microbial-derived biomarkers that can help in identifying individuals at high risk of diseases, ensuring the diagnosis of existing conditions, and monitoring the efficacy of treatment.

For example, a bacterial microbiota diversity analysis—used to check for differences in the strain and number of microbes in chronic kidney disease patients and healthy people—found that there was an abundance of Bacteroidetes and Proteobacteria and fewer Firmicutes as compared to a healthy individual. Consequently, this provided a marker for the diagnosis of CKD. Furthermore, there are some diseases that occur because of the translocation of microbes from their normal habitat into some other tissues, where they can trigger diseases by setting up an inflammatory cascade. For example, it was found in a study that lupus, an autoimmune disease, was caused because of the translocation of *Enterococcus gallinarum* from the gut to the liver and other tissues, which triggered an autoimmune response. These are two instances of the potential use of the human microbiome as a diagnostic tool.

Several firms are focusing on investigation in this area. Enterome SA, for example, raised USD 13.8 million in venture capital funding for developing tests that use the composition of gut bacteria to diagnose inflammatory and liver diseases. It also entered into an agreement with Bristol-Myers Squibb for the discovery and development of microbiome-derived biomarkers, drug targets, and bioactive molecules to be developed as potential companion diagnostics and therapeutics for cancer. This will greatly support the prospects of the human microbiome in clinical research.